

- Place the ballistic ball on the gun and push it to its locking position. The ball is now ready to be launched. (Check to see that the range is clear of any obstacle, especially a human being!)
- Mark the position right underneath the ball as $x = 0$ m.
- Launch the ball from its initial position and mark where the ball first hit the ground.
- Measure the horizontal distance (range) the ball has traveled, that is, between $x = 0$ and the point on the mark where the ball first hit the ground. Record the range in Table 4.7.1. Repeat this step 5 times and compute the mean range $\bar{x}$
- Compute the initial velocity v^0 of the ball. Use the equation

$$y = -\frac{gx^2}{2v_0^2}$$

4.6.2 Measuring the Range

- Assemble the ballistic gun as shown in Figure 4.2
- Incline the launcher to 20° , and lock it in place.
- Launch the ball and mark the where the ball first hit the ground. Measure the horizontal distance from the initial position of the ball ($x = 0$) to the point of the mark and record in Table 4.7.2. Repeat this step five times and then, take the average of the measured range record as x_{exp} .
- Compute the expected range, x_{theo} . Use the equation

$$R = x_{theo} = \frac{v_0^2 \sin 2\theta}{g}$$

- Compute the percentage error of the range using the equation:

$$\% Error_x = \left| \frac{x_{theo} - x_{exp}}{x_{theo}} \right| \times 100$$

Repeat steps 2 to 5 at different launching angles of 30° , 45° , 60° and 70° .

4.7 Data and Results

TABLE 4.7.1 Initial Velocity Determination				
Initial height	20cm			
Final Position (cm)				
x1	x2	x3	x4	x5
Mean distance, $\bar{x}$ (cm)				
Initial velocity, $\vec{v}_0$ (cm/s)				